CHAPTER 9
PRODUCT DOCUMENT

9.1 Product Overview

The **Personalized Learning Content Generator (PLCG)** is a web-based AI-powered learning assistant that generates structured, personalized educational content using Retrieval-Augmented Generation (RAG) and large language models.

The system supports:

- **Topic-based personalized explanations**
- **Real-life examples tailored to learner interests**
- **Conceptual questions and answers**
- **Flashcards for quick revision**
- **Adaptive learning through difficulty-based outputs**
- **Downloadable JSON and content export options**

PLCG is designed to improve learning outcomes by adapting explanations and questions to the learner's **age, interests, difficulty level, and preferred tone**.

9.2 Product Features

1. AI-Powered Content Generation Engine

Uses hybrid RAG + LLM pipeline for highly accurate and context-aware content:

- Summaries
- Explanations
- Examples
- MCQs
- Flashcards

- SBERT + FAISS for retrieval
- Fine-tuned Flan-T5 for structured content
- GPT-5-Nano (API) for high-quality outputs

2. Personalization Module

Automatically adjusts content based on:

- **Learner age**
- **Interests**
- **Preferred tone**
- **Difficulty level (easy/medium/hard)**

Ensures age-appropriate vocabulary, relatable examples, and customized explanations.

3. Multi-Format Output Generator

The system produces content in multiple educational formats:

- Structured notes
- Step-wise explanations
- Multiple-choice questions
- Flashcards (Q/A style)
- Raw JSON for developers

4. RAG-Based Knowledge Retrieval

- Semantic search using SBERT embeddings
- FAISS retrieval index for high-accuracy context selection
- Reduces hallucinations and improves factual consistency

5. Web Deployment (HTML/JS or Streamlit + FastAPI)

A clean, user-friendly, no-installation interface offering:

- Real-time content generation
- Output cards with copy/download actions
- Adjustable sliders for number of examples/questions
- Toggle-based selection of output formats

6. Export and Developer Tools

- Download JSON output
- Ready for API integration
- Ideal for LMS platforms or school apps

9.3 Product Architecture

The PLCG system consists of the following major layers:

• Data Processing & Preprocessing Pipeline

- Cleans user inputs
- Parses long text / chapter-based content
- Extracts key concepts using keyword extraction modules

• Retrieval Layer

- SBERT-based embedding generator
- FAISS index to fetch relevant chunks
- Ensures context-aware explanation and question generation

• Generative AI Layer

- Flan-T5 for summaries, explanations, structured notes
- GPT-5-Nano for high-quality MCQs and flashcards

- JSON-structured output generation
- Difficulty and personalization conditioning

• **Verification Layer**

- QA model to verify MCQ answers
- Distractor logic checks
- Factuality filters to reduce hallucination

• **API Layer (FastAPI)**

- Endpoints for:
  - /generate_gpt5
  - /notes, /summary, /mcq, /flashcards, etc.
- JSON-based communication between UI and backend

• **Front-End UI Layer**

- Built using Streamlit or HTML+CSS+JS
- Modules:
  - Topic input
  - Personalization settings
  - Output preview cards
  - Download & copy tools

9.4 Intended Users

The PLCG system is designed for:

- **Students** (school, college, competitive exam learners)
- **Teachers** (creating worksheets, tests, examples)
- **Parents** (simplified content for child learning)
- **Tutors & Coaching Centers**

- **Educational App Developers**
- **Content Creators / Instructional Designers**
- **Self-learners** exploring new concepts

9.5 Constraints

- **Model Accuracy Dependence**
  Depends on the quality of the retrieval index, fine-tuning data, and prompt design.
- **API Usage Limitations**
  Hosted LLMs (like GPT-5-Nano) may require API keys and incur usage costs.
- **Latency Variability**
  Response time may vary depending on:
  - Model size
  - Network environment
  - API throughput
- **Sensitive Content Filtering**
  Must avoid generating inappropriate or biased educational content without proper filtering.
- **Hosting Requirements**
  Continuous availability requires stable hosting for both backend and frontend.